NUS GRIP
Ambient Systems

Assignment #1

September 1, 2020

Introduction

Vision: create a safe, healthy, and productive indoor environment for all

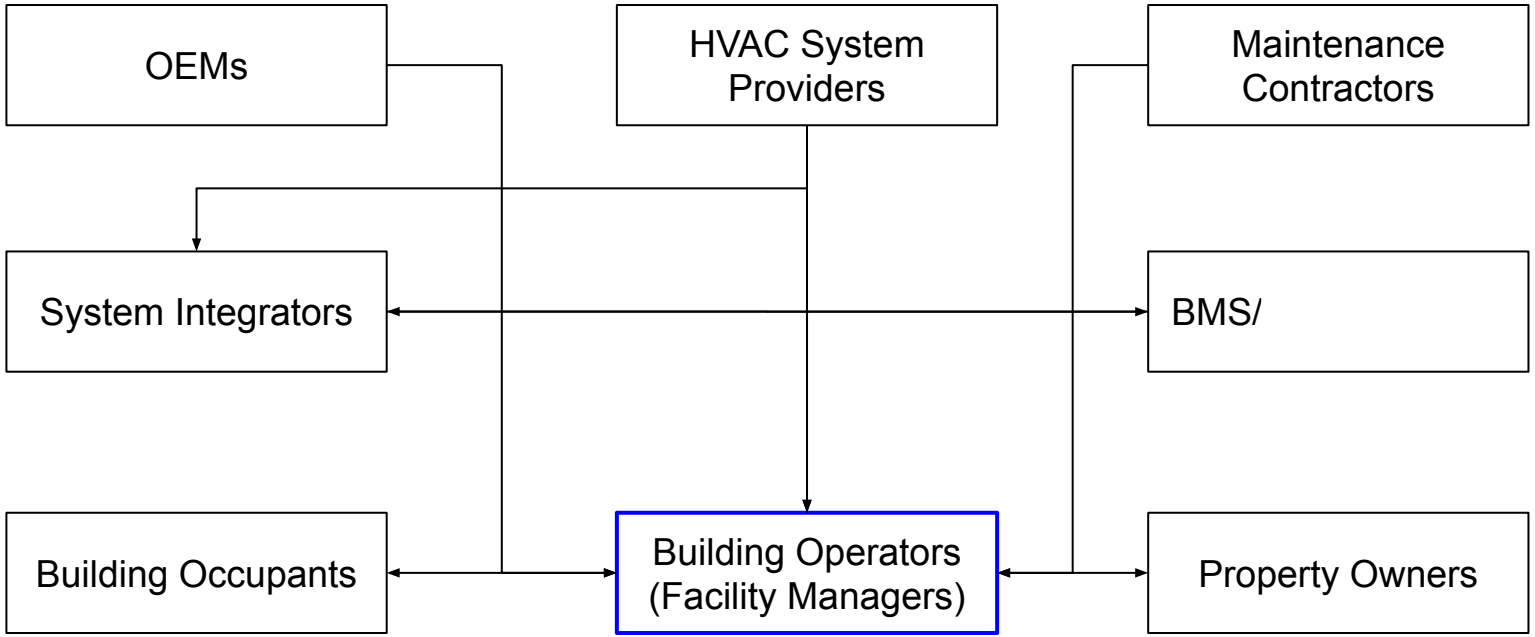
Mission: become a leading global provider of building management and automation systems

Goal: develop intelligent building automation solutions enabled by IoT and that delight property owners and building operators

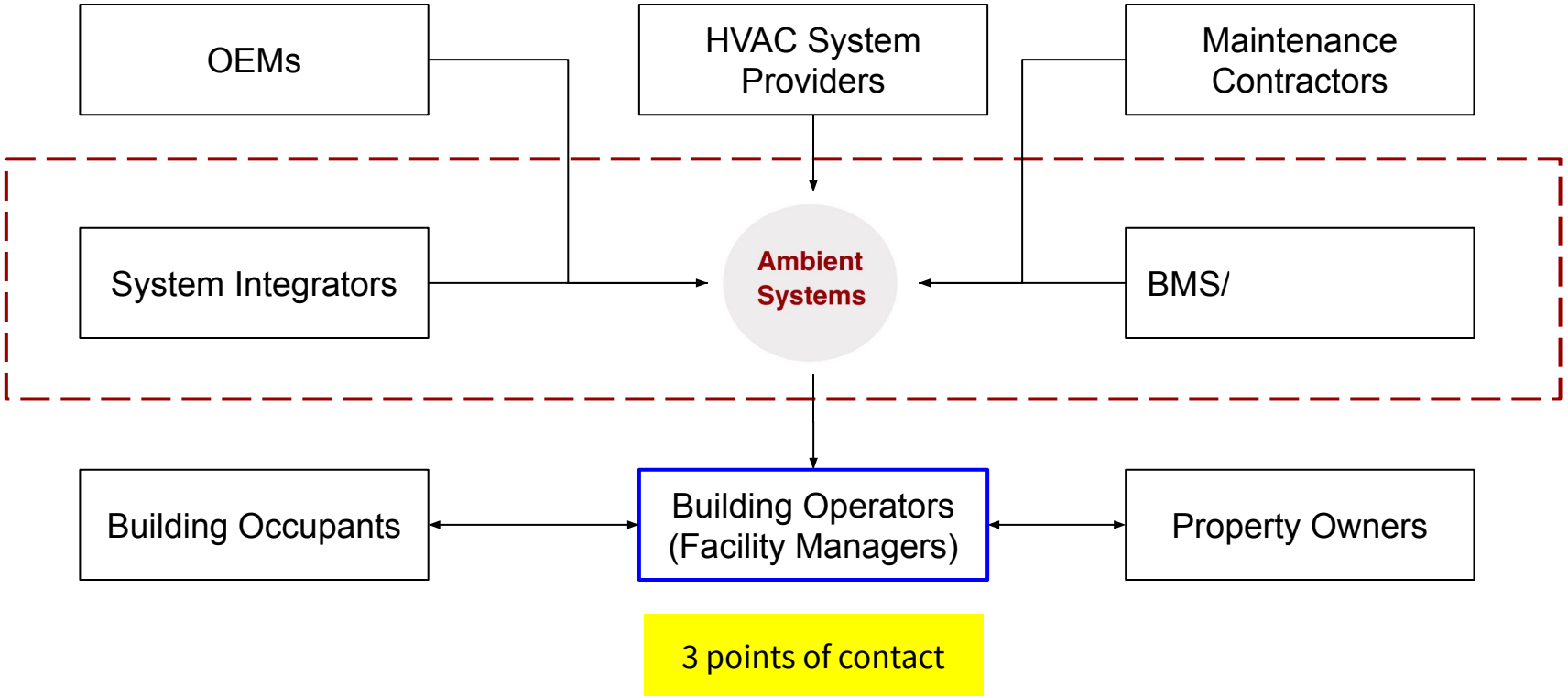
Section 1: Value Chain



Value Chain Analysis



Value Chain Analysis



Section 2: Paying Customer



Customer Segments

Building Segment	Value Proposition	Examples
Indoor Farms	Monitoring of relevant environmental parameters (e.g., light, air, water, and nutrients) Yield optimization (data analytics or ML)	Archisen Singrow Kingdom Fungi
Hospitals	Increase visibility into indoor air quality (IAQ) Reduce the spread of airborne hospital associated infections (HAI) (~10% of all HAI)	Parkway Pantai Singhealth National Healthcare Group
Commercial	Enhance productivity Ensure safety and compliance	Corporate offices Data centers
Museums	Protect staff and visitors Preserve invaluable artifacts	NUS Museum National Museum of Singapore

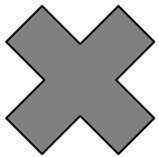
Why start with Indoor Farms?

- Clear need
- Rapidly growing
- Early adopters

$1M Customer

Annual Fee

$200/sq. m *



Property Size (Farm Area)

5,000 sq. m



$1 M Anchor Customer

* Pricing to be confirmed after further market validation and financial modelling.

Section 3: Value Proposition Canvas



Top Five Pain Points Across Segments

Top **five pain points** of property owners and building operators across segments:

1. Hard to find trained personnel to continuously manage building's indoor environment
2. No or minimal visibility into real-time environmental parameters
3. Outdated and incompatible legacy systems
4. Competing priorities (e.g., improve environmental parameters or reduce energy costs)
5. Evolving legal compliance and certification requirements

Key takeaway: building operators face similar problems despite differences in building use.

Value Proposition Canvas: Indoor Farms

Section 4: Market Sizing



Total Available Market (TAM)

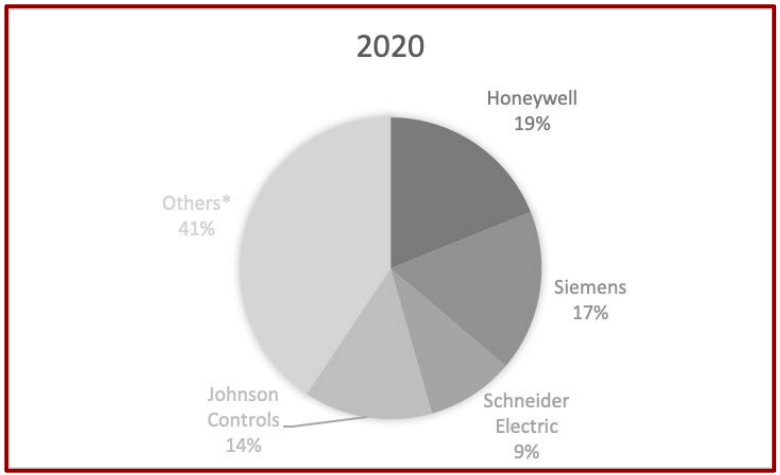
$4.74B

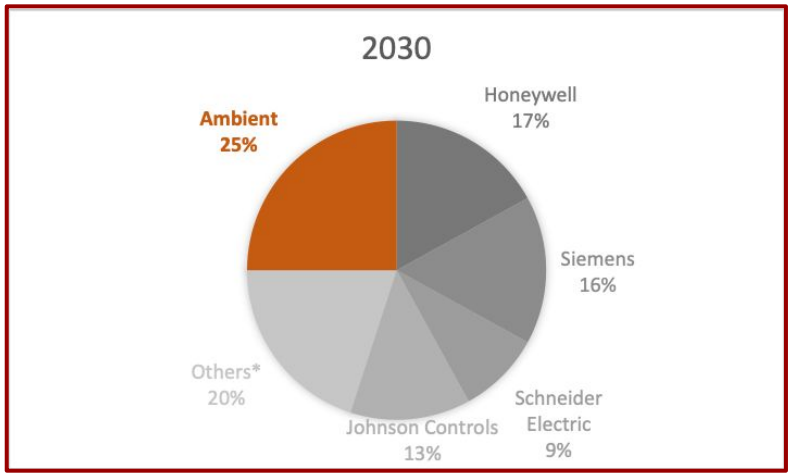
Total BAS Market: Revenue Forecast by Region, Global, 2017–2026

Year	Europe ($ Million)	North America ($ Million)	APAC ($ Million)	Latin America ($ Million)	MEA ($ Million)
2017	1,776.0	1,232.9	1,107.2	106.9	197.2
2018	1,823.5	1,269.9	1,149.2	111.0	203.5
2019	1,865.5	1,310.5	1,195.2	115.6	210.0
2020	1,878.5	1,321.3	1,210.2	117.7	212.9
2021	1,927.9	1,373.3	1,282.1	125.8	225.2
2022	1,963.6	1,420.8	1,340.7	131.5	234.1
2023	2,007.4	1,472.3	1,408.1	137.0	242.8
2024	2,059.8	1,523.4	1,483.4	143.2	253.3
2025	2,121.4	1,584.8	1,560.5	149.4	263.5
2026	2,184.8	1,655.2	1,646.5	155.4	274.9
CAGR	**2.3%**	**3.4%**	**4.7%**	**4.3%**	**3.9%**

Data from Frost & Sullivan

Serviceable Obtainable Market (SOM)

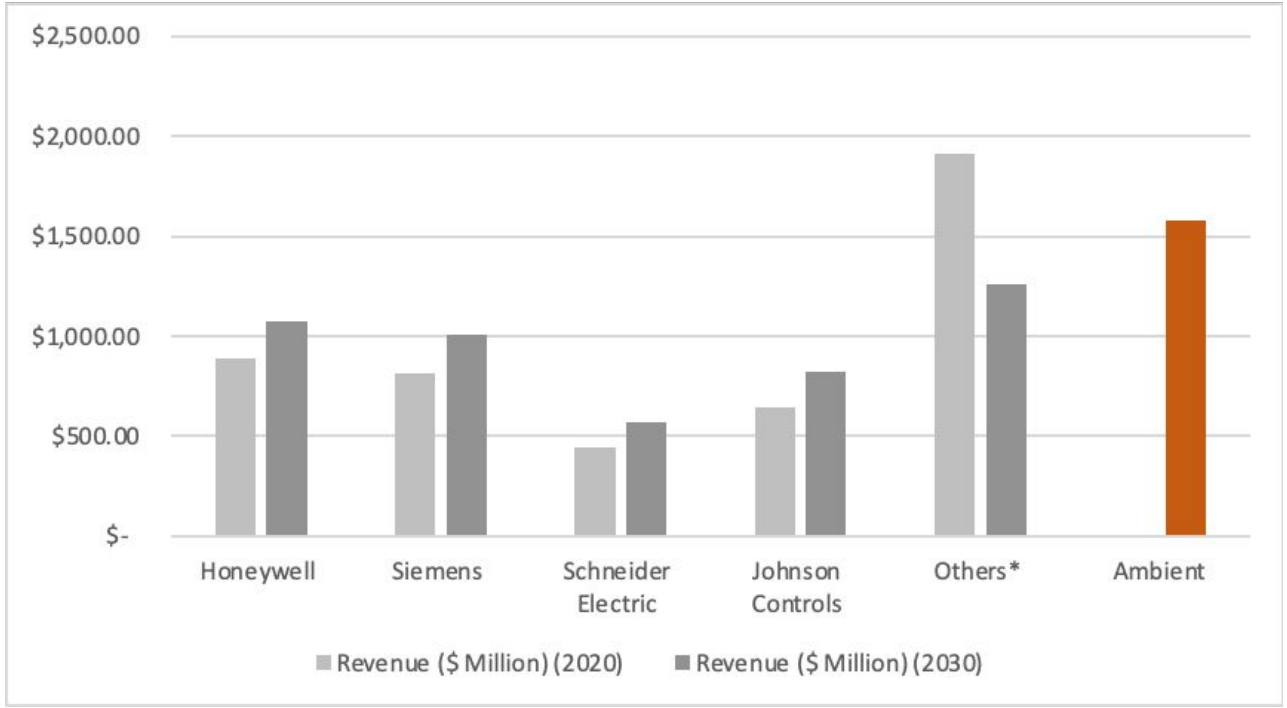




Data from Frost & Sullivan

Serviceable Obtainable Market (SOM)

$1.57B

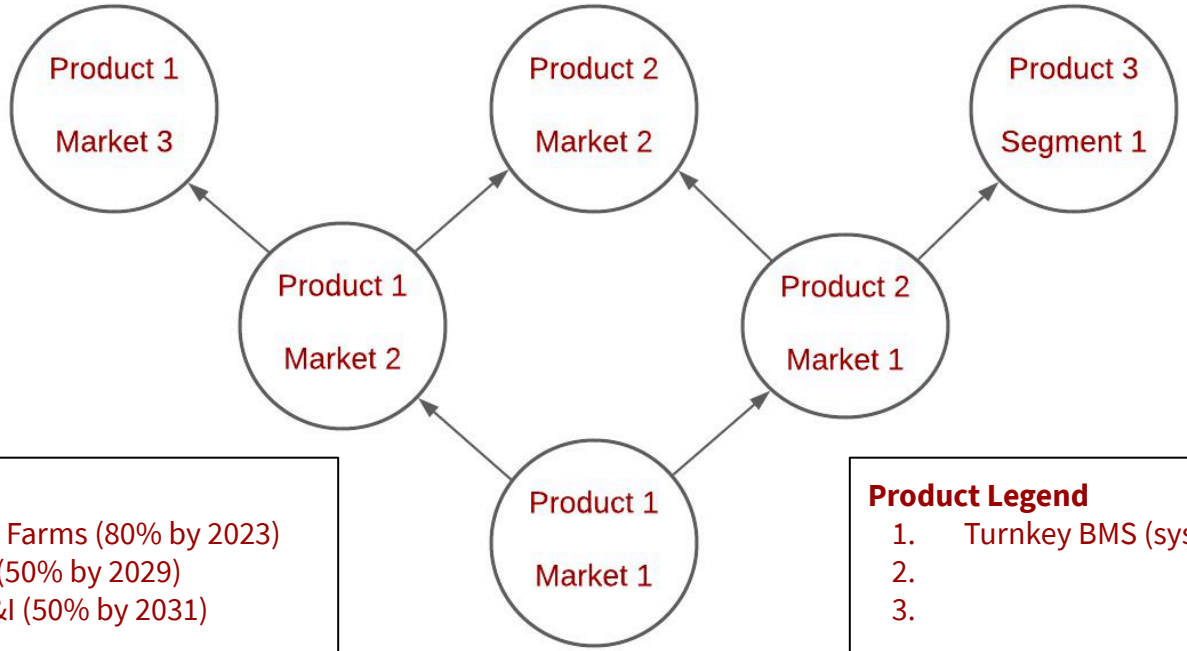


Data from Frost & Sullivan

Section 5: 3S Framework



Bowling Pin Diagram



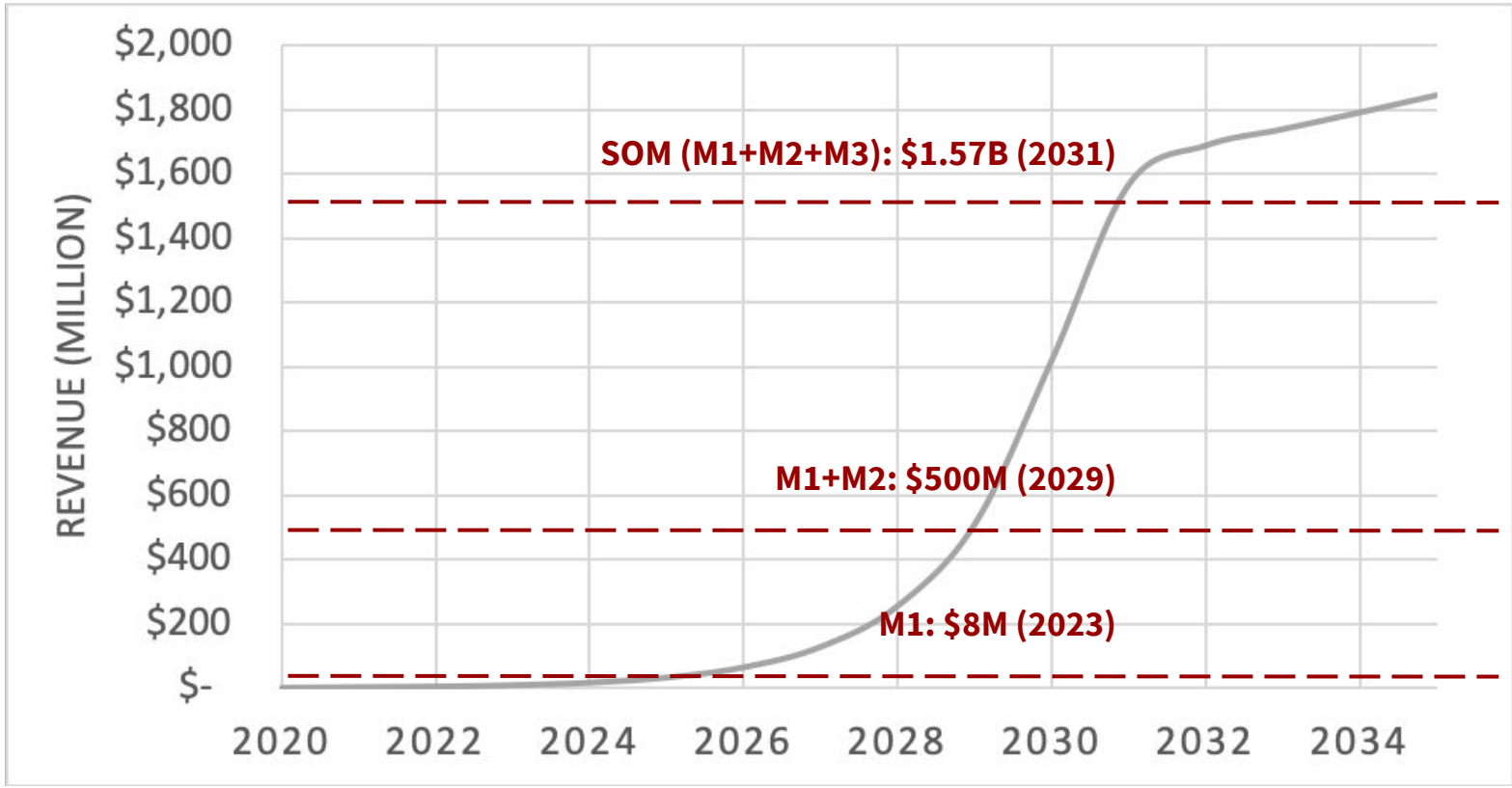
Market Legend

1. M1: SG Indoor Farms (80% by 2023)
2. M2: APAC C&I (50% by 2029)
3. M3: Europe C&I (50% by 2031)

Product Legend

1. Turnkey BMS (system integrator)
2. enabled BAS
3. enabled

3s Framework



Thank you

Appendix

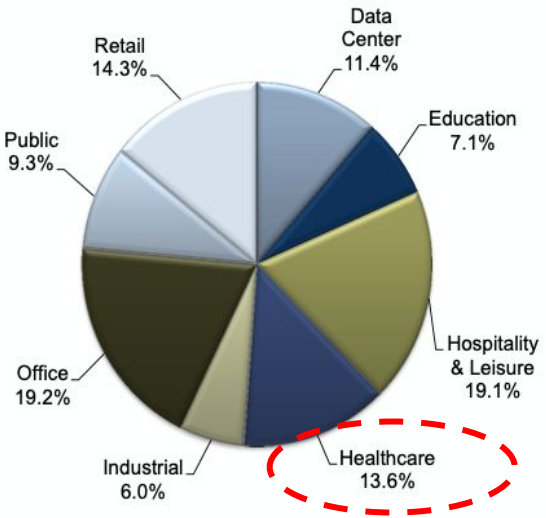


$1M Potential Customer (Hospitals)

Total BAS Market: Revenue Forecast by Region, Southeast Asia, 2015–2023

Year	Indonesia ($ Million)	Malaysia ($ Million)	Singapore ($ Million)	Thailand ($ Million)	Rest of Southeast Asia ($ Million)
2015	8.9	56.8	38.3	12.7	9.7
2016	9.6	60.2	39.8	13.9	10.3
2017	10.6	64.5	41.6	15.2	11.2
2018	11.9	69.6	43.7	17.1	12.1
2019	13.4	75.9	46.3	19.3	13.2
2020	15.3	83.5	50.0	22.2	14.4
2021	17.7	93.5	55.0	26.0	15.9
2022	20.7	106.6	61.6	30.6	17.6
2023	24.5	122.6	70.3	36.5	19.5
CAGR, 2018–2023	**15.6%**	**12.0%**	**10.0%**	**16.4%**	**10.0%**

BAS Market: Percent Revenue Breakdown by End User, Singapore, 2018



Healthcare Segment in SG = $50M * 13.6% = $6.8M

Data from Frost & Sullivan

$1M Potential Customer (Hospitals)

Parkway Pantai (in communication)	Singhealth	National Healthcare Group (NHG)	National University Health System (NUHS)
Mount Elizabeth Gleneagles Hospital Parkway East Hospital	KK Women's & Children's Hospital Singapore General Hospital Changi General Hospital Sengkang General Hospital	Tan Tock Seng Hospital Khoo Teck Puat Hospital Yishun Community Hospital	National University Hospital Ng General Hospital Alexandra Hospital

Four main healthcare groups manage 13 hospitals in Singapore that would benefit from our solution.

13 hospitals x $100k/year/hospital* = $1.3M/year in 2021

*We estimate we can provide an ongoing service (i.e., SaaS) at approximately $100k per hospital per year ($1 per 10 sq.ft.). Minimum contract period to be confirmed after further market validation and financial modelling.

Value Proposition Canvas (Hospitals)

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